

Sr. No.	Contents	Weekly Date
1	Implementation Progress – I	21/07/2025 to 26/07/2025 28/07/2025 to 02/08/2025 04/08/2025 to 09/08/2025
2	Implementation Progress – II	11/08/2025 to 16/08/2025 18/08/2025 to 23/08/2025 25/08/2025 to 30/08/2025
3	Implementation Progress – III	01/09/2025 to 06/09/2025 08/09/2025 to 13/09/2025 15/09/2025 to 20/09/2025
4	Testing, Results and Discussion	22/09/2025 to 27/09/2025 29/09/2025 to 04/10/2025 06/10/2025 to 11/10/2025
5	Report Writing, Conclusions, References / Bibliography	13/10/2025 to 31/10/2025

1. IMPLEMENTATION PROGRESS – I

Week 1: Review of Phase-1 Work and Initial GPS-Based Navigation Analysis

Week Duration: 21/07/2025 to 26/07/2025

Work Done:

At the beginning of Project Phase-2, a detailed review of the work completed in Phase-1 was carried out. By the end of Phase-1, the user interface design, ARCore integration, and basic floor detection functionality were already successfully implemented. The application was capable of detecting floor planes and placing virtual objects stably using ARCore anchors.

The focus of this week was to move beyond static AR visualization and begin working toward dynamic indoor navigation by tracking user movement. As an initial approach, GPS-based positioning was explored to determine whether it could be used to support navigation logic inside large indoor environments such as railway stations.

Several experiments were conducted to test GPS signal behavior inside buildings. The obtained readings were analyzed for accuracy, consistency, and reliability. It was observed that GPS signals were highly unstable indoors, with frequent signal loss and large positional errors caused by obstruction, multipath effects, and weak satellite visibility.

In parallel, ARCore floor detection was further refined by resolving issues related to plane detection instability and anchor alignment. Multiple test runs were performed to ensure consistent floor detection across different indoor locations. This ensured that the AR foundation from Phase-1 was stable and ready for advanced navigation features.

Based on experimental observations, it became evident that GPS-based navigation was not suitable for precise indoor positioning, leading to the need for an alternative approach in the next stage.

Outcome:

By the end of this week, the limitations of GPS for indoor navigation were clearly identified. The Phase-1 ARCore floor detection was confirmed to be stable, and the project direction was refined to explore sensor-based movement tracking instead of GPS.

Week 2: Literature Survey and Transition to Sensor-Based Navigation Approach

Week Duration: 28/07/2025 to 02/08/2025

Work Done:

This week focused on identifying a reliable alternative to GPS for indoor navigation through an in-depth literature survey and technical analysis. Research papers, journals, and existing indoor navigation systems were studied to understand commonly used movement tracking techniques in indoor environments.

Special attention was given to sensor-based navigation methods using smartphone hardware. The concept of Pedestrian Dead Reckoning (PDR) was studied, which estimates user position based on step count, direction, and distance traveled. Various step detection and heading estimation techniques using inertial sensors were reviewed.

The gyroscope sensor was identified as an effective component for detecting direction changes and orientation during movement. The speedometer and accelerometer were studied for their role in estimating walking speed and step count. Compared to GPS-based approaches, sensor-based techniques were found to be more accurate, infrastructure-free, and suitable for indoor environments.

Based on the survey findings, the system design was updated to replace GPS-based tracking with gyroscope and speedometer-based movement estimation. Preparations were made to integrate these sensors with the existing ARCore-based floor detection and navigation framework developed in Phase-1.

Outcome:

By the end of this week, a clear transition from GPS-based navigation to sensor-based step counting and direction estimation was finalized. The project was now ready to implement gyroscope and speedometer-based navigation integrated with ARCore in the following weeks.

Week 3: Firebase Authentication and User Management Implementation

Week Duration: 04/08/2025 to 09/08/2025

Work Done:

The primary focus of this week was to implement a secure and reliable user authentication system for the application. Firebase was selected as the backend service due to its robustness, scalability, and ease of integration with Android applications. Firebase services were integrated into the Android project using proper configuration files and dependencies.

Email and Password authentication was enabled through the Firebase Console. The user registration module was implemented to allow new users to create accounts using valid email IDs and secure passwords. Password strength validation rules were added to prevent the use of weak credentials and enhance application security.

The login functionality was developed to authenticate registered users and allow secure access to application features. Proper input validation and error handling were implemented to handle cases such as incorrect passwords, invalid email formats, and unregistered users. A Forgot Password feature was also implemented, enabling users to reset their passwords via email-based verification links provided by Firebase.

Extensive testing was carried out to validate the authentication workflow. Different scenarios such as successful registration, login failure, password reset, and session persistence were tested to ensure system reliability. User data was securely managed to prevent unauthorized access.

Outcome:

By the end of this week, a complete and secure authentication system was successfully implemented. The application now supports user registration, login, and password recovery, ensuring that only authorized users can access the indoor navigation features.

2. IMPLEMENTATION PROGRESS – II

Week 4: Manual Indoor Mapping and AR-Based Navigation Design

Week Duration: 11/08/2025 to 16/08/2025

Work Done:

During this week, the focus shifted toward designing and implementing a manual indoor navigation system using Augmented Reality. Since railway stations are complex indoor environments with multiple paths, platforms, and turns, a manual mapping approach was adopted to ensure accuracy and reliability. Detailed indoor maps were designed by defining fixed routes between source and destination points within the station.

Key checkpoints were identified along each route. These checkpoints represented important decision points such as turns, intersections, staircases, and entry or exit points. Each checkpoint was carefully positioned to guide the user step by step through the indoor environment. Turning points were clearly marked to ensure smooth directional changes during navigation.

ARCore anchors were implemented at each checkpoint to provide stable reference points in the physical environment. These anchors ensured that virtual navigation elements remained fixed relative to the real world. Directional arrows were then developed and displayed in the AR view to visually guide users from one checkpoint to the next.

Initial testing revealed issues such as arrow flickering, misalignment, and instability due to sensor noise and tracking limitations. These problems were addressed by applying pose smoothing techniques and improving anchor stability using ARCore's motion tracking capabilities. Multiple indoor test runs were conducted to verify arrow visibility, alignment accuracy, and navigation flow.

Outcome:

By the end of this week, a functional manual AR-based indoor navigation framework was successfully implemented. Users could visually follow AR arrows through predefined indoor routes, providing a reliable navigation experience without the use of GPS.

Week 5: Research Survey and Comparative Study of Indoor Navigation Technologies

Week Duration: 18/08/2025 to 23/08/2025

Work Done:

This week was dedicated to conducting an extensive research survey on existing indoor navigation and positioning technologies. The primary objective was to analyze current solutions, identify their limitations, and justify the selection of the proposed ARCore and sensor-based approach.

A detailed study was carried out on Wi-Fi fingerprinting techniques, which rely on signal strength measurements from access points. It was observed that while Wi-Fi-based systems are commonly used, they suffer from low accuracy (typically 4–8 meters) and require extensive infrastructure setup and continuous recalibration.

Bluetooth Low Energy (BLE) beacon-based systems were also analyzed. These systems offer slightly better accuracy than Wi-Fi but involve additional costs related to beacon deployment, battery maintenance, and calibration. Their performance is also affected by signal interference and environmental changes.

In addition, Inertial Measurement Unit (IMU)-based navigation techniques were studied, focusing on step detection and heading estimation using smartphone sensors. Several research papers related to AR-based navigation, sensor fusion, and pedestrian dead reckoning were reviewed. Comparative tables were prepared to evaluate accuracy, cost, scalability, and feasibility of each approach.

Based on the analysis, ARCore combined with IMU-based navigation was identified as the most suitable solution due to its infrastructure-free nature, high accuracy, and scalability.

Outcome:

The research survey provided strong technical and academic justification for selecting an ARCore and IMU-based indoor navigation approach. The findings formed a solid foundation for further implementation and research documentation.

Week 6: Sensor-Based Step Counting and Direction Estimation Implementation

Week Duration: 25/08/2025 to 30/08/2025

Work Done:

In this week, the focus was on implementing sensor-based positioning to estimate user movement indoors. Since GPS is ineffective in indoor environments, smartphone sensors were utilized to track user motion accurately. Step detection was implemented using the accelerometer sensor by applying peak detection algorithms on acceleration magnitude data.

The gyroscope sensor was used to estimate the user's heading direction and detect turning movements. This allowed the system to determine whether the user was moving straight or changing direction. Step count values were converted into approximate distance using stride length estimation models, enabling calculation of the distance traveled by the user.

Sensor data was integrated with the previously designed AR navigation checkpoints. This integration ensured that AR arrows dynamically updated based on the estimated user position. Manual checkpoints were also used to periodically reset position estimation and reduce cumulative drift caused by sensor noise.

Extensive testing was conducted by placing the phone in different positions such as hand, pocket, and bag. These tests helped validate the robustness and consistency of the step detection algorithm. Accuracy results showed reliable performance across different usage conditions.

Outcome:

By the end of this week, a reliable sensor-based positioning module was successfully implemented. Accurate step counting and direction estimation significantly improved indoor navigation performance, forming a critical component of the overall system.

3. IMPLEMENTATION PROGRESS – III

Week 7: Sensor Fusion Integration and Dynamic AR Navigation

Week Duration: 01/09/2025 to 06/09/2025

Work Done:

During this week, the focus was on integrating data from multiple smartphone sensors to enhance the accuracy and reliability of indoor navigation. Accelerometer, gyroscope, and speedometer data were combined to form a unified sensor fusion model. The accelerometer was used for step detection, the gyroscope for direction and orientation tracking, and the speedometer to estimate walking speed and refine distance calculation.

A sensor fusion algorithm was implemented to minimize noise and improve positional accuracy. By combining readings from multiple sensors, the system was able to produce more stable and reliable movement estimates compared to using individual sensors alone. This integration significantly reduced errors caused by sudden movements or sensor drift.

The AR navigation module was updated to dynamically adjust directional arrows based on real-time user movement. As the user walked, AR arrows continuously updated their orientation and position, ensuring accurate guidance at every step. Manual checkpoints were strategically used to reset position estimates, which helped reduce cumulative drift over long walking distances.

Extensive testing was carried out on long indoor paths simulating real railway station environments. These tests helped evaluate the consistency of navigation instructions and the stability of AR visuals across extended distances.

Outcome:

By the end of this week, an accurate and reliable sensor fusion-based navigation system was achieved. The integration of multiple sensors with ARCore significantly improved navigation precision and overall system robustness.

Week 8: Full Application Integration and User Interface Enhancement

Week Duration: 08/09/2025 to 13/09/2025

Work Done:

This week focused on integrating all developed modules into a single, cohesive Android application. Individual components such as authentication, station selection, route selection, sensor-based positioning, and AR navigation were connected into a smooth and logical application flow.

The user interface was enhanced to provide real-time navigation feedback. Information such as step count, remaining distance, direction indicators, and navigation alerts was displayed clearly to improve user experience. UI elements were optimized to ensure readability and ease of use during walking.

Several UI and navigation inconsistencies were identified during testing, including delayed updates, misaligned arrows, and screen transition issues. These issues were carefully debugged and resolved to ensure seamless navigation performance.

Multi-device testing was performed across different Android versions and hardware configurations to verify compatibility and stability. After successful integration and testing, a complete demonstration video was recorded showcasing the application workflow from login to destination navigation using AR guidance.

Outcome:

By the end of this week, the application was fully integrated, stable, and user-friendly. All system modules worked together seamlessly, and the application was ready for evaluation and demonstration.

Week 9: System Performance Evaluation and Comparative Analysis

Week Duration: *15/09/2025 to 20/09/2025*

Work Done:

In this week, the primary focus was on evaluating the overall performance and accuracy of the indoor navigation system. Extensive testing was conducted to measure step detection accuracy, positional error, and real-time responsiveness of the navigation instructions.

The obtained results were compared with existing indoor navigation approaches such as Wi-Fi fingerprinting and BLE beacon-based systems. Performance metrics including accuracy, infrastructure requirements, scalability, and cost were analyzed. The proposed system demonstrated superior performance with a positional error of only 2–3 meters, outperforming Wi-Fi and BLE-based methods.

All observations, accuracy measurements, and comparison results were carefully documented. Key learning outcomes from implementation and testing were summarized, highlighting the strengths of ARCore and sensor fusion-based navigation.

Based on the evaluation results, the system was refined and prepared for the final testing and documentation phase.

Outcome:

System performance was successfully validated through experimental evaluation and comparative analysis. The results confirmed the effectiveness and practicality of the proposed indoor navigation solution.

4. TESTING, RESULTS AND DISCUSSION

Week 10: Functional Testing and System Optimization

Week Duration: 22/09/2025 to 27/09/2025

Work Done:

During this week, comprehensive functional testing of the entire application was conducted. All modules including user authentication, indoor navigation, sensor tracking, and AR visualization were tested individually and collectively to ensure correct functionality.

Various test cases were designed to validate system behavior under different conditions. Minor bugs related to UI responsiveness, sensor synchronization, and AR rendering were identified during testing. These issues were promptly fixed to enhance system stability.

Step detection accuracy was measured by comparing actual steps taken with detected steps. Multiple test runs were conducted to ensure consistency and reliability. System parameters were fine-tuned to improve accuracy and reduce latency.

Outcome:

The system was thoroughly tested and optimized, ensuring accurate navigation performance and stable application behavior.

Week 11: Result Analysis and Performance Discussion

Week Duration: 29/09/2025 to 04/10/2025

Work Done:

This week focused on analyzing experimental results obtained from system testing. Accuracy metrics such as step detection rate, positional error, and navigation response time were evaluated in detail.

The performance of the proposed system was compared with existing indoor navigation techniques. Advantages such as infrastructure-free deployment, low cost, and high accuracy were discussed. Limitations related to sensor drift and environmental lighting conditions were also identified.

Result tables and graphs were prepared to visually represent performance comparisons. These results formed the basis for discussion and interpretation in the project report.

Outcome:

A clear and detailed performance evaluation was completed, providing strong evidence of the system's effectiveness.

Week 12: Final Testing Review and Documentation Completion

Week Duration: 06/10/2025 to 11/10/2025

Work Done:

In the final testing week, all test results and observations were reviewed and verified. Real-time performance and application stability were validated through repeated test runs.

Testing documentation was finalized, including test cases, results, and observations. A final evaluation summary was prepared, highlighting system strengths, accuracy achievements, and areas for future improvement.

This phase ensured that the system was fully ready for final report writing and academic submission.

Outcome:

The testing phase was successfully completed, and the system was confirmed to be stable, accurate, and ready for final documentation and evaluation.

5. REPORT WRITING, CONCLUSIONS & REFERENCES

Week 13: Project Documentation, Conclusion, and Academic Submission

Week Duration: 13/10/2025 to 31/10/2025

Work Done:

During this final phase of Project Stage-II, the primary focus was on complete documentation, result consolidation, and academic presentation of the work carried out throughout the project. A detailed project report was prepared following the prescribed university format and academic writing standards.

The report began with an introduction that clearly defined the problem of indoor navigation in large railway stations and highlighted the limitations of existing GPS-based systems. A comprehensive literature survey chapter was written, presenting a critical analysis of existing indoor navigation techniques such as Wi-Fi fingerprinting, BLE beacon-based systems, and sensor-based navigation. Relevant research papers, journals, and technical articles were cited to support the study.

The methodology and system design chapters described the overall architecture of the proposed system, including ARCore integration, sensor fusion techniques, checkpoint-based navigation, and application workflow. Detailed implementation chapters documented the development of authentication modules, AR navigation, sensor-based step counting, and system integration with appropriate explanations and screenshots.

The results and discussion section presented experimental findings, accuracy analysis, and performance comparisons with existing approaches. Graphs, tables, and observations were included to clearly demonstrate system effectiveness. Based on the obtained results, conclusions were drawn highlighting the success of the ARCore and IMU-based indoor navigation system. A future scope section was also included, suggesting enhancements such as automatic map generation, multi-floor navigation, and improved sensor fusion algorithms.

Proper references and bibliography were added using standard citation formats to acknowledge all research sources. In addition to the written report, a professional PowerPoint presentation was prepared, summarizing the project objectives, architecture, implementation, results, and conclusions. Finally, the research paper and presentation were submitted to a student research conference for evaluation and presentation.

Outcome:

By the end of this phase, Project Stage-II was successfully completed with full documentation, presentation materials, and academic submission. The project was finalized in a structured and professional manner, ready for evaluation and future extension.